Marine Fish Quality Monitor: IoT Supervision of a Fishing Boat's Fish Hold

Ice temperature, meltwater clogging and spoilage-gas monitoring with a summary display, data logging and a phone dashboard for the boat owner

Category	Level	Main board
Internet of Things	High school	NodeMCU ESP8266

1. ABSTRACT

Fishing boats keep their catch in an insulated hold packed with crushed ice. Three things ruin the catch at sea: the ice running out, meltwater that cannot drain because the outlet is blocked so the fish soak in dirty water, and fish that start to rot. The Marine Fish Quality Monitor watches all three with low-cost sensors on a NodeMCU ESP8266: a DHT11 for the hold temperature, an ultrasonic sensor for the height of the meltwater, an MQ-135 for the ammonia and amines given off by rotting fish and an MQ-4 for methane from decomposition. It logs every reading, works out a freshness score, shows only a short summary on a 4-digit TM1637 display and flashes a fault code on it when anything goes abnormal. The boat owner joins the board's own WiFi and sees everything live on a phone or laptop. A separate simulator page lets the whole system be demonstrated without sensors.

2. AIM OF THE PROJECT

- Keep the fish hold at ice temperature (0 °C) and warn when the ice is melting away.
- Measure the meltwater height and warn when too much water means the drain outlet is blocked.
- Detect the ammonia, amine and methane gases released by rotting fish.
- Log the data through the voyage, show a summary on the box and the full picture on the owner's phone or laptop.
- Provide a simulator so the system can be shown and tested without sensors.

3. PROBLEM STATEMENT AND NEED

Fish is one of the most perishable foods. On ice it stays fresh for days, but once the hold rises above 0 °C bacteria multiply quickly. Melted ice must drain away: if the drain clogs with scales and slime, the bottom layer of fish soaks in meltwater and spoils fast. Spoiling fish give off ammonia and amines such as trimethylamine (the 'fishy' smell), and deeper decay produces methane. The crew is busy fishing and the hold is closed, so these problems are often found only at the harbour, when the catch has already lost its price. A cheap monitor that watches the hold all the time and tells the owner immediately protects both the catch and the buyer.

4. COMPONENTS USED AND COST

Component	Model	Qty	Purpose	Approx. cost
Controller	NodeMCU ESP8266 (ESP-12E)	1	Reads sensors, logic, logging, hotspot, web server	Rs. 280
Display	TM1637 4-digit 7-segment	1	Summary values; flashes fault codes	Rs. 90
Temperature sensor	DHT11	1	Hold (ice) temperature and humidity	Rs. 110
Ultrasonic sensor	HC-SR04	1	Meltwater height on the hold floor	Rs. 90
Ammonia sensor	MQ-135 module	1	Ammonia / amine level from rotting fish	Rs. 160
Methane sensor	MQ-4 module	1	Methane trip from decomposition	Rs. 160

Component	Model	Qty	Purpose	Approx. cost
Indicator (optional)	Red 5 mm LED + 220 Ω	1	Alarm light	Rs. 5
Resistors	1 kΩ, 2 kΩ	1 set	Ultrasonic echo divider	Rs. 5
Power	5 V 1-2 A USB supply / power bank	1	Supply (MQ heaters need 5 V)	Rs. 150
Total				Rs. 1,050

Table 1: Bill of materials (approximate local prices)

5. CIRCUIT CONNECTIONS

The NodeMCU has only one analog input (A0). The MQ-135 uses it, because the ammonia level is shown as a number; the MQ-4 uses the digital output (DO) of its comparator module, set with its potentiometer, as a gas yes / no signal. The HC-SR04 echo is a 5 V signal, so it goes through a 1 kΩ / 2 kΩ divider (or the sensor is powered from 3.3 V). The TM1637 is mounted upside down in the box, so the firmware rotates every digit by 180°.

Part pin	NodeMCU pin	Notes
TM1637 CLK / DIO	D5 / D6	VCC from VIN (5 V)
DHT11 DATA	D4	VCC 3.3 V
HC-SR04 TRIG / ECHO	D7 / D8	ECHO through 1 kΩ / 2 kΩ divider
MQ-135 AO	A0	VCC 5 V, DO not used
MQ-4 DO	D1	VCC 5 V, AO not used
Red LED (optional)	D0	through 220 Ω to GND

Table 2: Wiring (full sheet in CONNECTIONS.md)

6. SYSTEM OVERVIEW

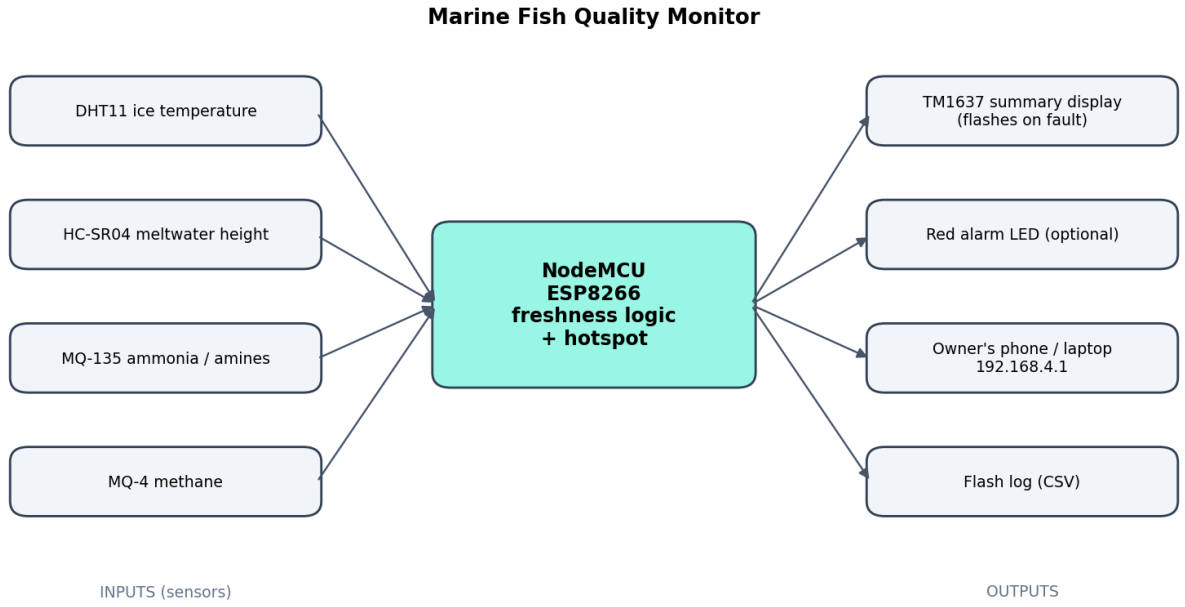


Figure 1: Block diagram

7. HOW I DID THIS (METHOD)

The ultrasonic sensor is fixed under the lid looking straight down. Meltwater height = (distance to the empty floor) - (measured distance); the median of the last five pings ignores splashes from the rolling boat. The DHT11 sits in the ice between the fish, and the two gas sensors sit in the air above the catch. The MQ-135

reading is averaged over 8 samples; in clean air it reads about 420 on this board. Gas alarms are ignored for the first 60 s while the heaters warm up, and the display shows a spinning segment with a countdown.

A freshness score starts at 100 and loses points for temperature above 0 °C (6 per degree), meltwater above 2 cm (10 per cm), MQ-135 readings above clean air (1 per 7 counts) and an MQ-4 trip (25). It is graded FRESH, GOOD, CHECK or SPOILED. The TM1637 normally slides through the temperature, water height, ammonia level and score, with a small water-ripple animation between rounds. When anything is abnormal it flashes a code - ICE, drAn, nH3 or CH4 - followed by the value, and the red LED blinks. A missing sensor shows E-dH, E-US or E-nH and appears as NO SENSOR on the dashboard, never as a made-up value. Every minute a CSV row is written to the board's flash; the dashboard gives the board the real time from the phone, so log rows get real dates.

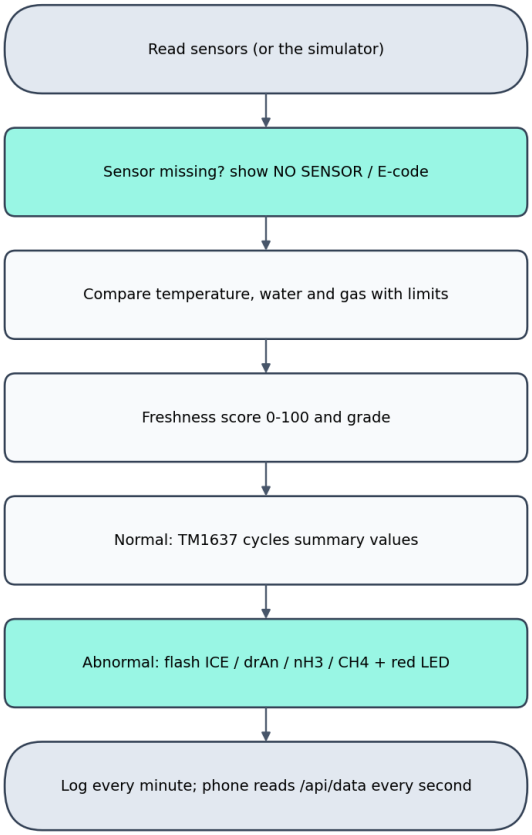


Figure 2: Program flow

8. SIMULATOR

A separate simulator page runs on any laptop without the board. The full dashboard sits on one side and sliders on the other: ice temperature, meltwater height, MQ-135 level, humidity and the MQ-4 trip, plus one-tap scenes (fresh catch, ice melting, drain blocked, rotten fish). A 'voyage story' mode plays a 4-minute trip through every alarm. The simulator uses exactly the same rules and limits as the firmware. The real board's page has no simulator controls and shows only live sensor readings; for exhibitions the board can still play the voyage story on its own display.

9. TESTING AND RESULTS

The graph below is a replay of the firmware's own simulator and alarm formulas over one voyage story. Every alarm fires in the right phase and clears afterwards; the freshness score falls to 0 at the worst point (181 s, rotting fish without ice) and recovers with fresh ice.

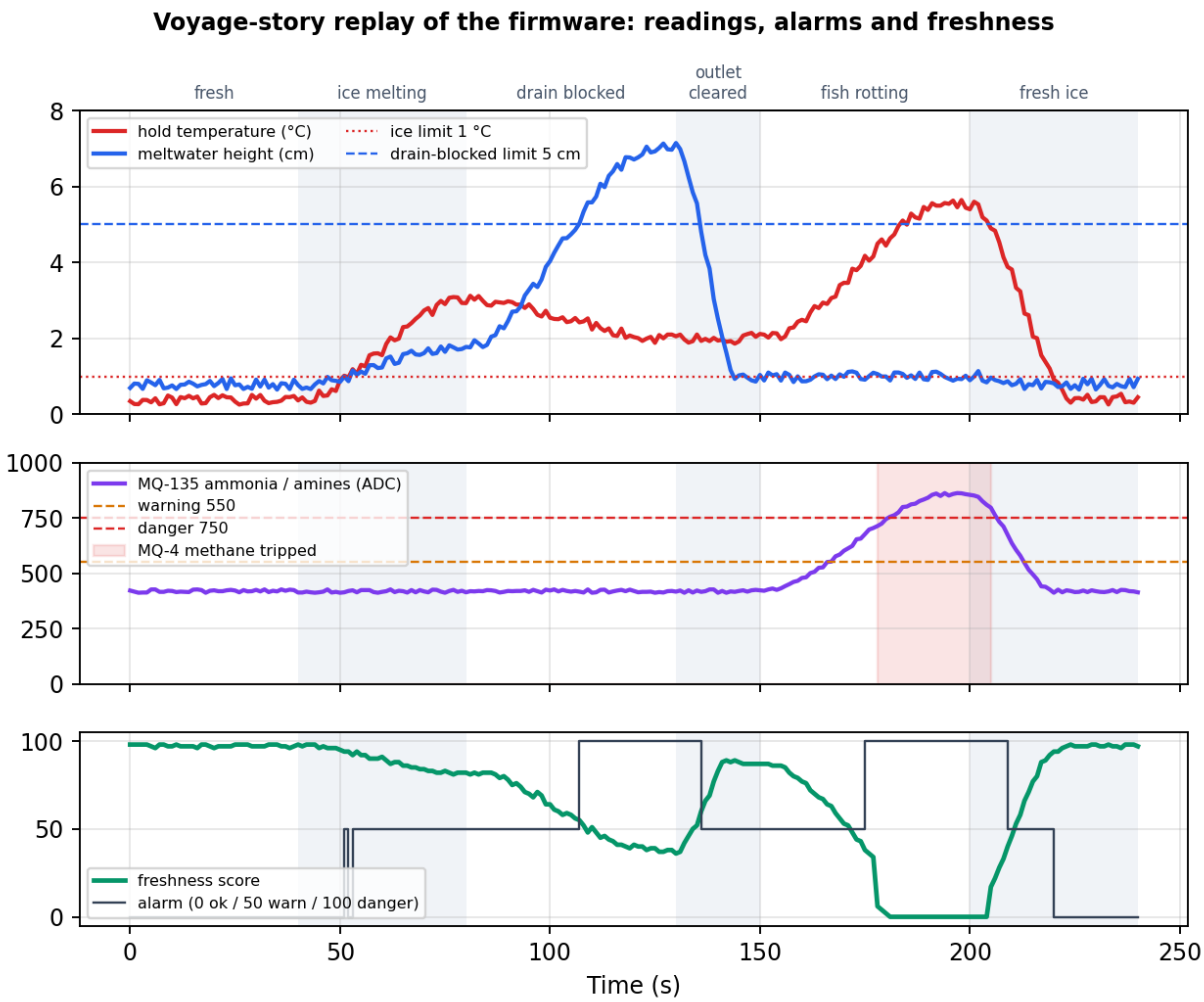


Figure 3: Readings, alarms and freshness during the simulated voyage

Phase of the voyage	What the sensors saw	Box display	First alarm
Fresh catch on ice (0-40 s)	0.4 °C, 0.8 cm water, MQ-135 ~420	cycles 0°C / H 1 / nH41 / Fr97	none
Ice melting (40-80 s)	temperature climbs to 3 °C	flashes ICE	51 s
Drain outlet blocked (80-130 s)	water rises to 7.1 cm	flashes drAn	86 s
Outlet cleared (130-150 s)	water drains back to 1 cm	ICE only	-
Fish rotting (150-200 s)	5.5 °C, MQ-135 up to ~863, MQ-4 trips	flashes nH3, CH4, ICE	166 s / 178 s
Fresh ice (200-240 s)	all values back to normal	summary again	cleared

Table 3: What the monitor reported in each phase

On the real board the MQ-135 read about 400-420 in clean room air, well below the 550 warning limit. The firmware compiles for the NodeMCU (32 % of flash, 40 % of RAM), and the dashboard and simulator were tested in phone- and laptop-sized browsers.

10. APPLICATIONS

- Fishing boats on multi-day voyages.
- Insulated fish boxes at harbours and auction halls.
- Fish transport vans and cold storage rooms.
- Fish markets checking the freshness of the catch.

11. ADVANTAGES AND LIMITATIONS

- Watches ice, meltwater and spoilage gases together for about Rs. 1,050.
- Owner sees it on any phone; no internet or app needed.
- Logs the whole voyage; simulator for demonstration and training.
- Limitation: the DHT11 cannot read below 0 °C; a DHT22 or DS18B20 would be more precise.
- Limitation: MQ sensors give relative readings and must be calibrated in clean air; the MQ-4 only gives yes / no here because the NodeMCU has a single analog input.

12. FUTURE IMPROVEMENTS

- Waterproof DS18B20 probe for exact ice temperature.
- An ADS1115 add-on to read both gas sensors as levels.
- GSM / LoRa link so the owner on shore gets alerts.
- Automatic drain pump when the meltwater is too high.

13. CONCLUSION

The Marine Fish Quality Monitor turns a closed fish hold into something the owner can watch from a phone. It keeps a record of the voyage, raises a clear alarm when the ice runs out, the drain blocks or the fish start to spoil, and shows its readings honestly - helping fishermen land a fresher, more valuable catch.

14. REFERENCES

- FAO - Ice in fisheries (Fisheries Technical Paper 331).
- Hanwei Electronics - MQ-135 and MQ-4 datasheets.
- Aosong - DHT11 datasheet.
- ESP8266 Arduino core documentation (ESP8266WebServer, LittleFS).